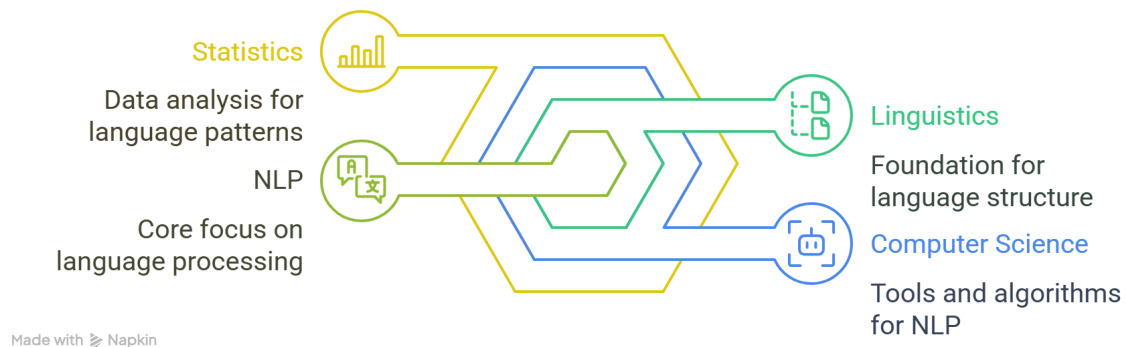


Introduction to Natural Language Processing (NLP)

What is NLP?

Natural Language Processing (NLP) is the area of computer science and artificial intelligence that focuses on enabling machines to understand, interpret, generate, and interact using human (natural) language. It sits at the intersection of linguistics, computer science, and statistics, and covers a wide range of tasks from simple text cleanup to complex language understanding and generation.

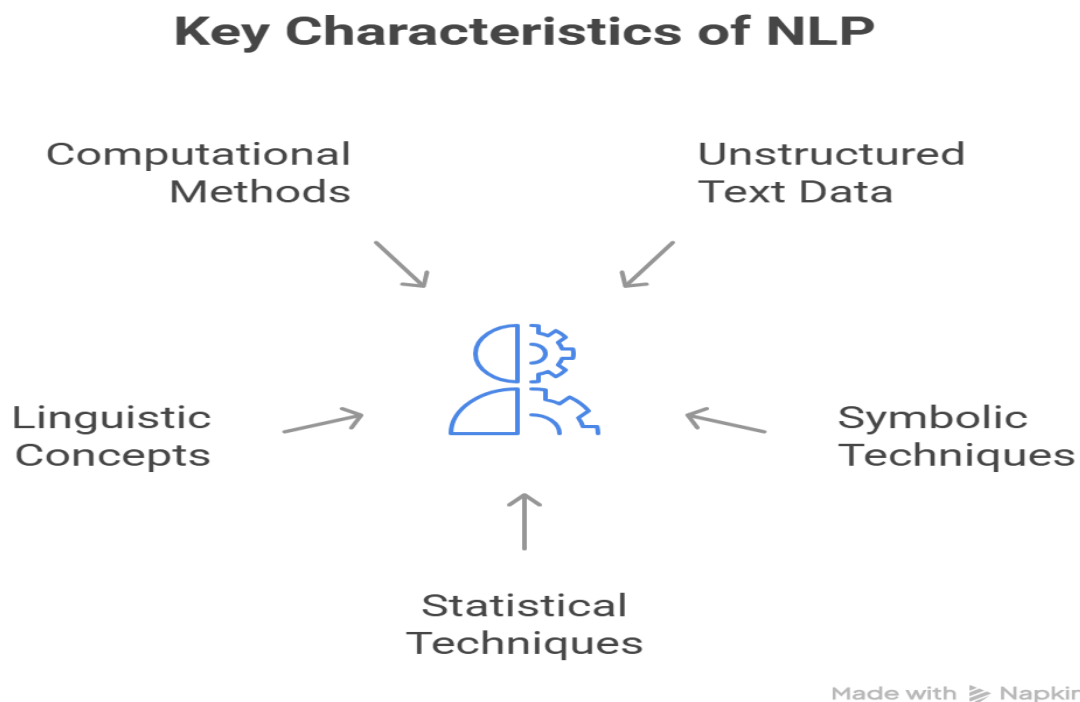
NLP and Its Interdisciplinary Nature



At a high level, NLP answers questions such as:

- What does this sentence mean?(understanding)
- Which words are named entities (people, places, organizations)? (information extraction)
- Is this review positive or negative? (classification)
- How should a machine respond to a human utterance? (dialogue and generation)

Key characteristics of NLP:



- Deals with messy, unstructured text data (and often speech).
- Requires both symbolic (linguistic rules) and statistical (data-driven) techniques.
- Combines linguistic concepts (syntax/semantics/pragmatics) with computational methods.

Importance & Why NLP Matters

Language is the primary medium through which humans share information, ask questions, make decisions, and record knowledge. Because of that, NLP unlocks tremendous value across industries:

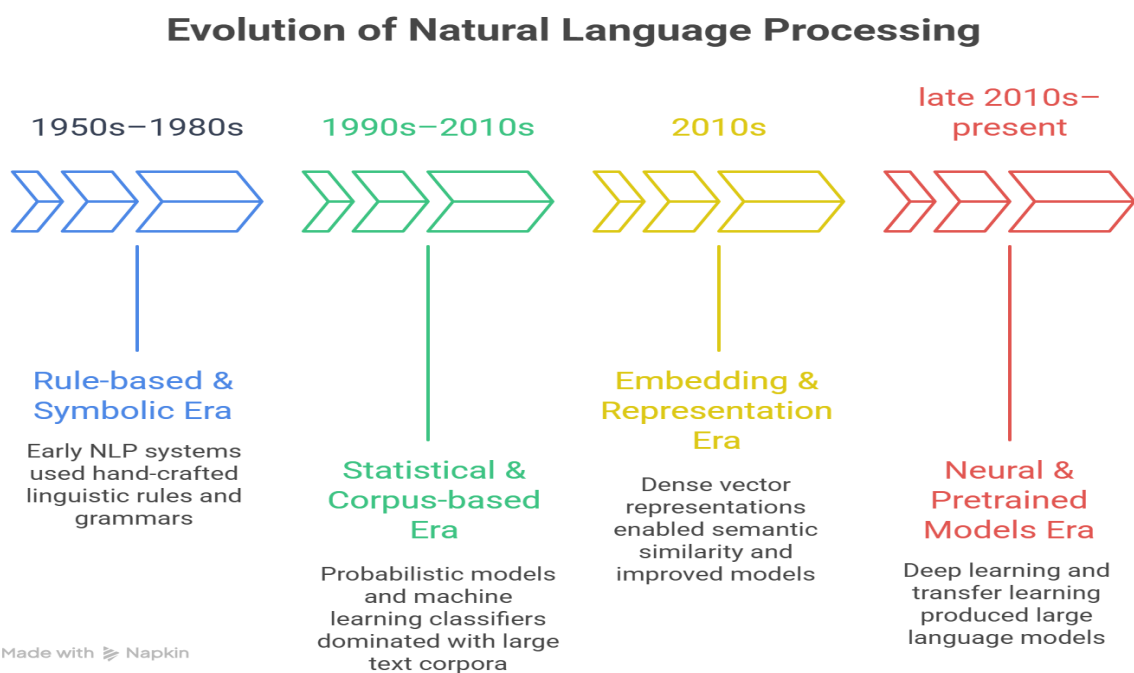
- **Search & Information Retrieval:** Better search results, semantic search, and question answering improve user experience in search engines and enterprise knowledge bases.

- **Customer Support & Automation:** Virtual assistants, chatbots, and automated ticket triaging scale support operations and reduce human workload.
- **Business Intelligence:** Sentiment analysis and topic detection provide insights from customer feedback, social media, and reviews.
- **Accessibility & Inclusion:** Text-to-speech, speech-to-text, and language simplification enable access for people with disabilities and language barriers.
- **Healthcare & Science:** NLP extracts clinical facts from medical notes, helps literature review, and supports medical coding and triage.
- **Localization & Translation:** Machine translation and multilingual NLP break language barriers in global applications.

Because textual data is everywhere (emails, documents, social media, reports), NLP converts this rich but unstructured information into actionable insights.

History & Evolution (concise timeline)

NLP has evolved through several broad paradigms:



1. Rule-based & Symbolic Era (1950s–1980s)

- Early systems used hand-crafted linguistic rules and grammars. Classic examples include pattern-matching programs and early parsing systems.

2. Statistical & Corpus-based Era (1990s–2010s)

- With large text corpora and better computing, probabilistic models (n-grams, HMMs, CRFs) and machine learning classifiers (SVM, logistic regression) became dominant.

3. Embedding & Representation Era (2010s)

- Dense vector representations of words (Word2Vec, GloVe) enabled measuring semantic similarity and improved downstream models.

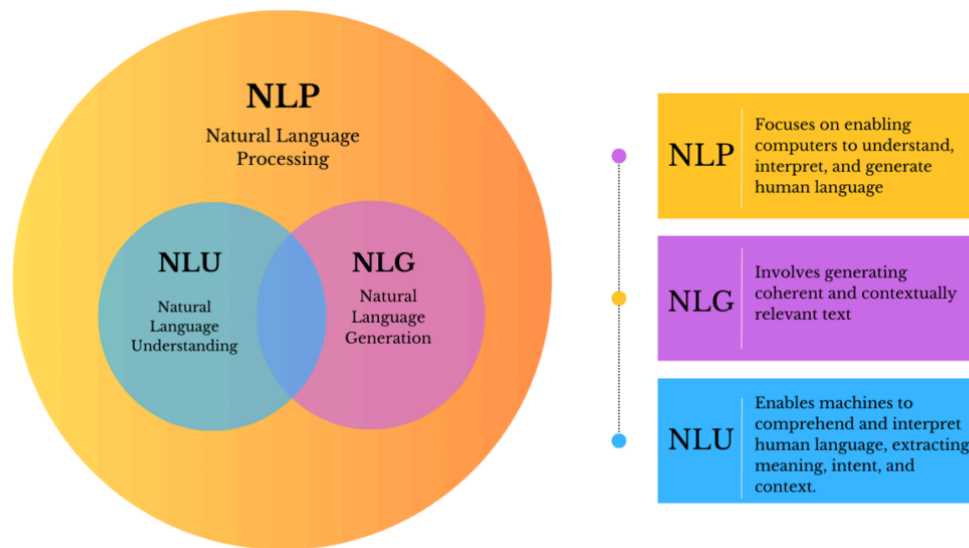
4. Neural & Pretrained Models Era (late 2010s → present)

- Deep learning architectures, pretraining on massive corpora, and transfer learning produced large language models. (Note: detailed model architectures and transformer internals are covered in your Deep Learning notes.)

This evolution reflects a move from hand-coded linguistic knowledge toward data-driven representations and then toward large-scale pretrained models.

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NLP vs NLU vs NLG — What's the difference?



Although these terms are related, they describe different aspects of language technology:

NLP (Natural Language Processing)

- Umbrella term for computational techniques that process natural language. Includes both understanding and generation tasks.
- Examples: tokenization, parsing, feature extraction, classification, translation, summarization.

NLU (Natural Language Understanding)

- Focuses on interpreting and extracting meaning from text or speech.
- Emphasizes analysis: intent detection, slot filling, semantic parsing, sentiment understanding, entity recognition, coreference.
- Example: From “Book a flight to Delhi next Friday,” NLU extracts intent (`book_flight`) and slots (`destination: Delhi`, `date: next Friday`).

NLG (Natural Language Generation)

- Focuses on producing fluent natural language from structured data or internal representations.
- Examples: template-based generation ("Hello, {name}!"), data-to-text generation (generating weather reports from measurements), and dialogue response generation.

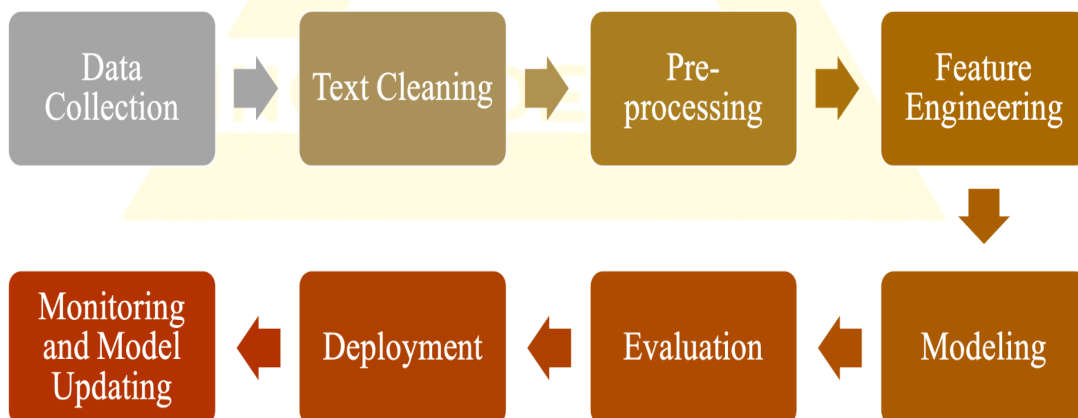
Simple way to remember:

- NLP = the entire pipeline and toolbox.
- NLU = "What does the text mean?"
- NLG = "What should the machine say/write?"

Many real systems combine NLU and NLG - e.g., a chatbot first uses NLU to understand the user, then uses NLG to respond.

NLP Pipeline Overview (step-by-step)

NLP Pipeline



A typical NLP pipeline organizes processing into stages. Not every project needs every stage — but the pipeline gives a logical order of operations.

1. Data Collection

- Gather raw textual data from sources such as social media, emails, documents, news, chat logs, or speech-to-text systems.
- Quality and quantity of data directly affect model performance.

2. Text Cleaning

- Remove unwanted characters, symbols, HTML tags, or emojis.
- Handle cases like multiple spaces, spelling errors, special characters.
- Objective: make the text consistent and noise-free.

3. Pre-processing

- Tokenization: splitting sentences into words/tokens.
- Stopword removal: eliminating common but meaningless words ("is", "the", "at").
- Stemming & Lemmatization: reducing words to base/root form.
- Normalization: lowercasing, handling slang, abbreviations.

4. Feature Engineering

- Convert cleaned text into numerical representations that ML/DL models can understand.
- Approaches:
 - Bag of Words (BoW)
 - TF-IDF

- Word embeddings (Word2Vec, GloVe, FastText)
- Contextual embeddings (BERT, GPT)

5. Modeling

- Train machine learning or deep learning models on extracted features.
- Examples:
 - Logistic Regression, SVM (traditional ML).
 - RNN, LSTM, GRU (sequential models).
 - Transformer models (BERT, GPT).

6. Evaluation

- Assess model performance with appropriate metrics:
 - Classification tasks: Accuracy, Precision, Recall, F1.
 - Translation/Summarization: BLEU, ROUGE.
 - Language modeling: Perplexity.

7. Deployment

- Integrate the trained model into real-world applications (chatbots, search engines, sentiment analysis systems).
- Provide APIs or user interfaces for interaction.

8. Monitoring & Model Updating

- Continuously track model performance in production.
- Detects issues like data drift (new slang, evolving context).
- Retrain/update the model with fresh data to maintain accuracy.

Note: Many modern systems interleave these stages (for example, representation learning can replace manual feature engineering). Because you are keeping deep model details in your Deep Learning notes, this pipeline focuses on conceptual stages and classical techniques.

Key Challenges in NLP (with examples & mitigation ideas)

Natural language is messy. Below are the most common challenges you should know and how practitioners typically address them.

6.1 Ambiguity

- **Definition:** A sentence or expression that admits multiple interpretations.
- **Examples:** "I saw the man with a telescope." (Who has the telescope?)
- **Mitigation:** Use syntactic parsing, semantic context, world knowledge, or probabilistic models to prefer likely interpretations.

6.2 Polysemy & Homonymy

- **Polysemy:** A single word has multiple related meanings (e.g., "bank" = financial institution vs river bank).
- **Homonymy:** Two unrelated words that share spelling/pronunciation.
- **Mitigation:** Word sense disambiguation using context windows, sense inventories, or distributional similarity. (Deep contextual models accomplish this more effectively; covered in Deep Learning notes.)

6.3 Slang, Jargon & Evolving Language

- **Problem:** Social media, domain-specific writing, and spoken language use non-standard spellings, abbreviations, and new words.
- **Mitigation:** Domain-specific corpora, normalization modules, subword tokenization to handle novel words, and continual model updates.

6.4 Context Dependence

- **Problem:** Meaning often depends on broader discourse, user intent, and background knowledge (coreference, ellipsis, implicature).
- **Example:** "John left because he was tired." (Who was tired?)
- **Mitigation:** Coreference resolution, discourse-level models, contextual features, and dialog state tracking in conversational systems.

6.5 Data Sparsity & Low-Resource Languages

- **Problem:** Many languages and domains lack large labeled corpora.
- **Mitigation:** Transfer learning, cross-lingual embeddings, data augmentation, and rule-based/knowledge-driven approaches.

6.6 Irony, Sarcasm & Figurative Language

- **Problem:** Literal word meanings differ from intended meaning, making automatic sentiment or intent detection hard.
- **Mitigation:** Use broader context, pragmatics, metadata (emojis, punctuation), and specialized classifiers trained on annotated corpora.

6.7 Domain & Distribution Shift

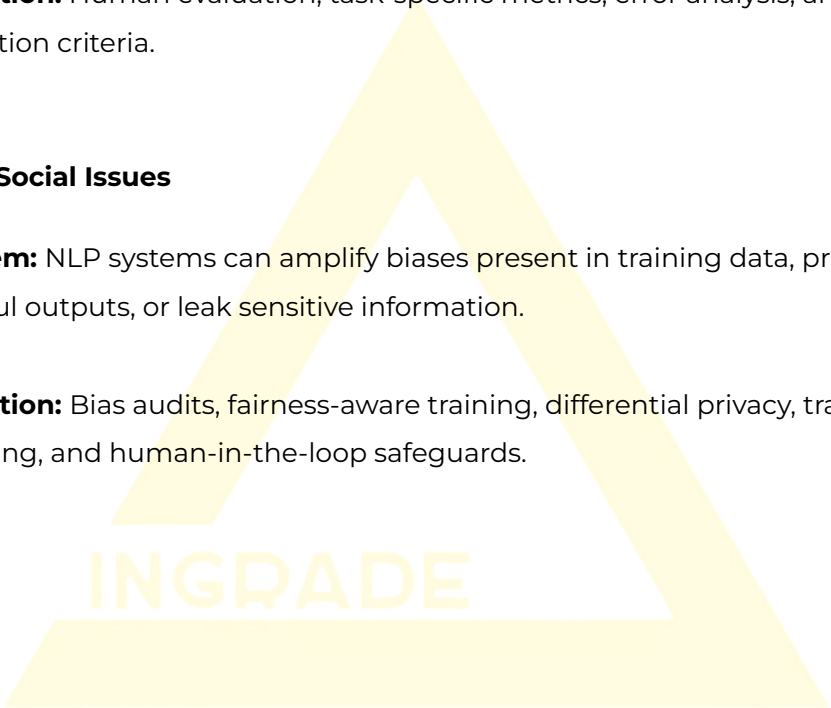
- **Problem:** Models trained on one domain (e.g., news) can fail on another (e.g., tweets).
- **Mitigation:** Domain adaptation, fine-tuning on in-domain data, and careful feature selection.

6.8 Evaluation Challenges

- **Problem:** Automatic metrics for generation (BLEU, ROUGE) do not fully capture human judgments; ambiguous labeling can make supervised evaluation noisy.
- **Mitigation:** Human evaluation, task-specific metrics, error analysis, and multiple evaluation criteria.

6.9 Ethical & Social Issues

- **Problem:** NLP systems can amplify biases present in training data, produce harmful outputs, or leak sensitive information.
- **Mitigation:** Bias audits, fairness-aware training, differential privacy, transparency in reporting, and human-in-the-loop safeguards.



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Glossary (quick reference)

- **Token:** A minimal unit (often a word or subword) produced by tokenization.
- **Corpus:** A large collection of text used for training or analysis.
- **Vocabulary:** Set of unique tokens recognized by a model.
- **OOV (Out-of-Vocabulary):** Tokens unseen during training.
- **Lemma:** The canonical/dictionary form of a word.
- **Stem:** The crude root form produced by a stemmer.
- **POS (Part-of-Speech):** Grammatical category such as noun, verb, adjective.
- **NER (Named Entity Recognition):** Detecting entities like PERSON, LOCATION, ORGANIZATION.

